

Conglomerate Mergers: Effects on Growth and Competition

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January 5, 2026

Abstract

We examine the effects of conglomerate mergers on firm growth and the competitive landscape using an agent-based model. We provide evidence that resource sharing within conglomerates acts as a risk mitigation strategy in multiplicative growth processes, providing incentives for conglomerate formation. Our model shows that even small levels of resource sharing can significantly alter market dynamics, leading to more balanced market share distributions and increased firm mobility. Contrary to concerns about market concentration, our results suggest that conglomerate formation can reduce both static and dynamic measures of market concentration.

JEL Classification Codes: C63, D21, D43, L11, L40

Keywords: Conglomerate Merger, Competition, Growth Optimization, Risk, Agent-Based Modeling

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1 Introduction

Conglomerative mergers are important to many fields of economic inquiry, ranging from industrial organization [3, 15, 16], international trade [5, 26, 19], to finance [18, 6] and even management studies [2, 1]. However, unlike horizontal or vertical mergers, whose drivers have been studied thoroughly, the determinants of conglomerative mergers have not received nearly as much attention. Formally, it defines the merging of two businesses that operate in different markets and are not part of the same value chain. Many of these mergers take place across borders, which is why they are also known as complex foreign direct investment [5]. We note that market and product extension mergers [12] may also be partially motivated by the pooling mechanism we identify, since gaining additional growth streams aids the diversification process. By isolating pure conglomeration, we identify the independent contribution of resource pooling to merger incentives, which may operate alongside other motivations in more complex merger types. The determinants of such processes remain both manifold and elusive. Among those brought forward are economies of scale, and increasing market power [21], taxation [10] or alignment with management salary risk preferences [1]. One overarching theme, however, is the reduction in risk. For example, [29] points to a reduction in the standard deviation of profits by merging two firms operating in independent markets. Despite the fact that the standard deviation is neither technically, nor methodologically a good measure of risk [35], it provides valuable insights into the determinants of conglomerative mergers. In this context [21] introduces the concept of growth maximization as a reason for conglomeration, where diversifying across firms yields an increased rate of growth for the owning entity. Considering this in light of recent ideas by [23] offers a new path forward. In particular, the notion of firm growth rate optimization dynamically through time by reducing losses seems both novel and established. Under this framework an agent (here a firm) searches for the most cost-effective means to reduce negative impacts on their growth rate. [1] propose that mergers act to avoid disastrous outcomes, almost as some form of insurance, provided that the premium paid in terms of manage-

ment costs is low enough. Adding more companies to a portfolio to reduce losses works best, of course, when these companies' outcomes are maximally independent. [3] presents this thought by proposing that mergers happen at different stages of the firms' lifecycle, thus increasing the diversification potential.

Just as unsolved remains the question of the impact that such mergers have on the competitive landscape. While [38] finds that post acquisition companies grow at an above industry average rate, [16] and [15] do not detect any consequences that would increase market concentration. The results in [38] seem to depend on the acquiring company to financially support the acquired company as a necessary condition of out performance. Naturally, there is a limit to the size of a company as management cost, such as the coordination of all constituent businesses, increase [9]. This provides a natural drag on the effectiveness of conglomeration as a means to increase the growth rate. The direction of the effect on competitiveness is of course relevant to policy-makers, wishing to optimize consumer welfare.

We will offer evidence to help propel the answer to both of these questions. First, we demonstrate that under multiplicative dynamics, resource sharing within conglomerates is positive-sum rather than zero-sum, providing a novel theoretical foundation for conglomerate formation. Second, we distinguish between portfolio diversification and what we term 'true diversification' through pooling, showing that the latter offers superior growth rate optimization. Third, we provide simulation evidence that these incentives, if acted upon, would reduce rather than increase market concentration. Section 2 provides an introduction into growth rate optimization under multiplicative dynamics and their relevance in trying to understand competitive processes. Section 3 presents the formal model, as well as the assumptions implied in it. Continuing in Section 4, we show the results of the simulation, and introduce and discuss the metrics used in our arguments. Finally, we put these into context in Section 5 before concluding in Section 6.

2 Background

Conventional economic models had great success explaining the formation of vertical and horizontal mergers. Despite this, conglomerative mergers have escaped such reasoning so far. Many explanations are based on cost sharing arguments [21], whereby firms split up expenses among themselves. One point often made against such an explanation is that this is in effect zero-sum, given that one firm's gain is another's loss. However, this critique only holds under the assumption of additive growth. In a setting of multiplicative dynamics, where the next state s_{t+1} of some variable (e.g. market share, reputation, size) is dependent on the previous one s_t [14, 34], the objective is to maximize the growth rate g , the geometric average of periodic increments $\delta_t \in (0, \infty)$, sometimes also referred to as the Compound Annual Growth Rate in investment circles

$$g = \left(\prod_{t=1}^T \delta_t \right)^{\frac{1}{T}} \quad (1)$$

These increments are the changes to the variables in question, such as the change in sales, the effect of marketing campaigns on the way consumers perceive a firm, etc. Crucially, the lower increments are, the greater their impact on the growth rate. While an increment of $\delta_t = 0.2$ already requires the next one to be $\delta_{t+1} = 1.25$ to compensate it, one of $\delta_t = 0.5$ requires δ_{t+1} to be 2. This asymmetry in influence of losses over gains grows exponentially and thus follows a logarithmic function of the increments' impact on the growth rate as seen in Figure 1. From the point of view of avoiding the smaller levels of δ_t or increasing them across the board, sharing cost and even revenues in a conglomerate becomes a positive-sum game, as the reduction in the gain of the more successful firms is overcompensated by the increase in gain of the others. This holds true regardless of whether all firms make profits, only some do, or even none do. Suppose there are two firms in a conglomerate and absolute increments defined as $\Delta_{it} = \delta_{it}s_{it} - s_{it}$ [34]. Furthermore, without loss of generality suppose $\Delta_{it} > \Delta_{jt}$ and a sharing

level of x . Given that $\frac{\partial^2 \log(x)}{\partial^2 x} < 0$, it is clear that $\log(\Delta_{it}) - \log(\Delta_{it} - x) < \log(\Delta_{jt} + x) - \log(\Delta_{jt})$.

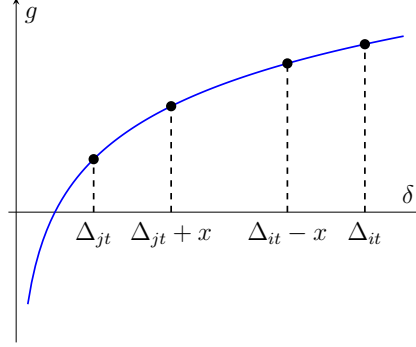


Figure 1: Impact of growth increments on the compound growth rate.

One can think of this as raising the base of the state s_{t+1} such that the next growth increment δ_{it+1} impacts a greater absolute value. For example, assuming a multiplicative process consisting of normally distributed increments with mean μ and standard deviation σ , the time average growth rate, the average increment, converges to

$$g = \mu - \frac{\sigma^2}{2} \quad (2)$$

In addition to the mean μ , the fluctuation of the increments $\frac{\sigma^2}{2}$ has an outsized impact on the growth rate, as the negative fluctuations outweigh the positive ones even though they are symmetric. We can easily demonstrate this since even when gaining and losing at the same rate the overall outcome will be net negative δ_t : $(1 - \delta_t)(1 + \delta_t) = 1 - \delta_t^2 < 1$. This dynamic is sometimes referred to as *variance drain* [20] or *volatility drag* [32]. Thus, as outlined, reducing the magnitude of fluctuations is the largest lever to increase the growth rate.

Again, we can link this to sharing, where conglomerate formation increases g for all firms within the entity. Under multiplicative dynamics, sharing can be shown to be growth optimal. Following [24], with N firms, whose growth increments δ_i are independently drawn from the same distri-

bution $\mathcal{N} \sim (\mu, \sigma)$, the *sharing* or *pooling* of their increments increases their individual growth rates such that

$$g = \mu - \frac{\sigma^2}{2N} \quad (3)$$

As the number of firms N in a conglomerate increases, the drag imposed by the fluctuations decreases. In the extreme case of $N \rightarrow \infty$ each firm is able to realize the ensemble average μ with every increment. Of course there are limits to how big a firm and thus a conglomerate can grow, given that management cost, especially for coordination increase [9]. Any increase in conglomerate size will increase management cost, while at the same time increasing the growth rate for all firms included. Thus, one can imagine this kind of cooperation as a hedge, or an insurance against unfavorable growth increments, the premium for which is the cost of coordination. Clearly, there is a trade-off involved, and a conglomerate size, though certainly varying, which optimizes it. This concept is somewhat related to that of diversification, as applied to financial portfolios. However, in contrast to the latter, pooling also improves the growth rates of the entities that make up the portfolio, while offering even greater growth prospects than diversification to the portfolio itself. It is, in this sense, *true diversification*, which we define as a cooperative arrangement that increases the time-average growth rate of each constituent unit under multiplicative dynamics, rather than merely reducing variance at the portfolio level

3 Model

3.1 Growth Process

Agent-based models (ABMs) have become increasingly useful in the context of economic modelling [4]. However, they have not been employed to model conglomerative merging behavior of firms. To investigate how growth rate optimization influences the incentives in conglomerate formation, we set up

an ABM with M markets that are uncorrelated between each other, each containing an amount of N firms. Firms face random multiplicative growth that is unique to their market. Each increment δ_{it} is drawn from a normal distribution $\mathcal{N}(\mu_m, \sigma_m)$ with positive ensemble average μ_m . In this framework each firm's periodic state, which can be seen as its size, is multiplicatively linked to its previous one.

$$s_{it+1} = s_{it}\delta_{it} \quad \text{with} \quad \delta_{it} \sim \mathcal{N}(\mu_m, \sigma_m) \quad (4)$$

We model markets such that the mean and the standard deviation of the distribution over their growth increments are positively correlated, meaning that a higher mean μ_m comes with a higher standard deviation σ_m . As explained in section 2, in multiplicative processes one is not just maximizing for the ensemble average μ_m , but for the growth rate g . Hence, a higher scale σ_m decreases the growth rate, through the volatility tax.

Each firm thus realizes an absolute period gain of $\Delta_{it} = s_{it-1}\delta_{it} - s_{it-1}$. For any firm not part of a conglomerate this will also be their period profit $\Pi_{it} = \Delta_{it}$. Firms in conglomerates C pool an amount $\alpha\Delta_{it}$, which flows into the conglomerate's period pool $\Omega_{ct} = \sum_{i \in c} \alpha\Delta_{it}$, with $\alpha \in [0, 1]$. The growth process is multiplicative at the shock level but additive at the accounting level. Firms experience multiplicative shocks to size, but pooling operates on realized absolute gains, which are naturally additive. This reflects the institutional reality that internal capital markets redistribute realized resources rather than growth rates themselves. As conglomerates grow in size, maintaining them becomes ever more difficult as management costs increase. We model these using four different functional forms:

$$\Phi(O_c) = b_{0p}|O_c|^{b_{1p}} \quad (5)$$

$$\Phi(O_c) = b_{0l} + b_{1l}|O_c| \quad (6)$$

$$\Phi(O_c) = b_{0q} + b_{1q}|O_c| + b_{2q}|O_c|^2 \quad (7)$$

$$\Phi(O_c) = b_{0e}e^{b_{1e}|O_c|} \quad (8)$$

The parameters b are management cost scaling hyperparameters for the power law (p), linear (l), quadratic (q), and exponential (e) cost functions. The literature on management diseconomies provides theoretical foundations [9, 39] but limited empirical guidance on the precise functional form of coordination costs. [39] suggests that diseconomies arise from communication distortion, bureaucratic insularity, and incentive limits, but these mechanisms could plausibly generate linear, convex, or even initially concave cost structures depending on organizational form and industry context. [8] find empirically that diseconomies and scale economies tend to offset each other, making the functional form difficult to identify directly. Given this theoretical and empirical ambiguity, we test four functional forms representing the main possibilities: power law (initially concave), linear, quadratic (convex), and exponential (strongly convex). Rather than treating this as purely a robustness check, we treat the cost structure as a parameter of interest: while the direction of pooling effects is robust across specifications, the magnitude varies substantially, which has implications for optimal conglomerate size. We calibrate the cost function parameters such that all four specifications yield equivalent costs at a conglomerate size of 40 members, which represents a mid-range conglomerate in our setup with $M \in \{50, 100, 150\}$ markets. We also vary these parameters around these baseline values across simulations to ensure our results are not driven by a particular calibration choice. Once computed, this cost factor scales the cost according to conglomerate size $\sum_{i \in c} s_{it}$ such that the pool size is given by

$$\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c) \sum_{i \in c} s_{it} \quad (9)$$

where the set of markets a conglomerate operates in is defined as $O_c \subseteq M$.

Lastly, this pool will be shared equally, determining a member's profit by $\Pi_{it} = (1 - \alpha)\Delta_{it} + \frac{\Omega_{ct}}{|O_{c_i}|}$. Consequently, any firm's next state is given by $s_{it+1} = s_{it} + \Pi_{it}$.

3.2 Merging and Splitting

At the beginning of every time step t each firm $i \in M \times N$ gets a chance to choose another with probability p . If successful, it will select a random other firm j , among the remaining $(M \times N) - 1$ firms. We allow conglomerates to only operate one firm per market, meaning it can have a maximum of M members. Hence, conglomeration is only *feasible* if firm j and i :

1. don't operate in the same market ($m_i \neq m_j$)
2. are not part of a conglomerate that operates in a market in which the other firm, or the conglomerate which it is part of operates in ($O_{c_i} \cap O_{c_j} = \emptyset$)

We assume that if both firms are in conglomerates that these will merge as well. After confirming *feasibility*, merging firms determine *desirability*, meaning that they would both benefit from joining forces. They do so by comparing the realized pool Ω size over the last l periods, against a "synthetic" pool $\hat{\Omega}$ size if they had merged.

Specifically, firms compute $\hat{\Omega}$ by applying the pooling formula 9 to their combined historical increments $\{\Delta_{it}, \Delta_{jt}\}_{t=t_l}$ over the lookback period, using the management cost that would apply to the combined entity. This assumes firms have access to each other's historical performance data, which could be interpreted as the due diligence process preceding merger negotiations. If $\hat{\Omega} > \Omega$ the merger will be implemented.

After an initial l periods each firm that is part of a conglomerate evaluates its performance within it by comparing its growth rate $g_{i(t-l):t}$ being part of the conglomerate against the potential rate $\hat{g}_{i(t-l):t}$ it had realized, had it remained outside. Upon being worse off, a firm exits the conglomerate and continues to operate alone until it gets the chance to join another conglomerate. The full model can be seen in algorithm 1.

Algorithm 1: Model flow

```
for  $t$  in  $T$  do
  foreach  $i \in N \times M$  do
    Merge with  $j$  if  $(m_i \neq m_j)$  and  $(O_{c_i} \cap O_{c_j} = \emptyset)$  and  $\hat{\Omega} > \Omega$ 
  foreach  $c \in C$  do
     $\Omega_{ct} = \sum_{i \in c} \alpha \Delta_{it} - \Phi(O_c) \sum_{i \in c} s_{it}$ 
    foreach  $i \in c$  do
       $\Pi_{it} = (1 - \alpha) \Delta_{it} + \frac{\Omega_{ct}}{|O_c|}$ 
    foreach  $i \notin c$  for all  $c \in C$  do
       $\Pi_{it} = \Delta_{it}$ 
    foreach  $i \in c$  for all  $c \in C$  do
      Exit  $c$  if  $\hat{g}_{i(t-l):t} > g_{i(t-l):t}$ 
```

4 Results

We implemented the model outlined in the previous Section in Python with extensive robustness testing. For each of our hyperparameters, to be found in Table 1, we chose a set of values to analyze whether they influence the key results. In addition to these, we also vary the parameters of our cost functions. For each of the four cost function types we ran a counterfactual altering one of the hyperparameters. Within each counterfactual we gradually increase the parameter of interest, the sharing rate α , by 0.02 to investigate the response of our model. To gain statistical confidence in our results, each counterfactual is run 50 times per level of α in a Monte Carlo setting, which leaves us with a total of 70,200 simulations.

Given the high-dimensionality of our results, we estimate several panel regressions, controlling for the counterfactual heterogeneity to isolate the effect of our variable of interest α on the question outlined in Section 2. Table 2 shows the impact of a change in pooling on the per period number of average members per conglomerate, number of conglomerates, mergers, and exits (defined as a firm leaving a conglomerate). Importantly, we estimate a quadratic model to capture any non-linear effects, which are clearly relevant

Parameter	Value
M	$\{50, 100, 150\}$
N	$\{50, 100, 150\}$
T	10000
l	$\{10, 50, 100\}$
p	$\{0.01, 0.05, 0.1\}$
Φ	l, e, p, q
α	$[0, 0.5]$

Table 1: Model Parameters and their values

	(1) Avg. Members	(2) Num. Conglomerates	(3) Mergers	(4) Exits
α	11.45*** (0.01)	4603.32*** (3.74)	226.25*** (0.36)	225.09*** (0.37)
α^2	-12.87*** (0.02)	-7857.29*** (7.23)	-483.93*** (0.70)	-482.74*** (0.71)
Obs.	14040000	14040000	14040000	14040000
R ²	0.70	0.63	0.35	0.34

Note: Standard errors in parentheses.

*p<0.1; **p<0.05; ***p<0.01

Table 2: Panel regression results showing how pooling affects conglomerate size and count

and statistically significant, meaning that pooling increases all metrics up to a point before impacting them negatively. When looking at the effect sizes of periodic mergers and exits, we see that increasing α increases mergers slightly more than exits indicating stable formation of conglomerates. Additionally, we observe that increasing α produces more conglomerates with fewer members on average, indicating stronger market fragmentation. These effects are, however, not homogenous across cost structures. Figure 4 shows that power law cost scaling leads to comparatively larger, but fewer conglomerates. It also produces merging and exit patterns that are much less dynamic than those of rest of the cost structures. Since costs scale initially slower in the power law case, conglomerates can form more reliably, lowering the number of exits and consequently also the number of firms able to join new ones.

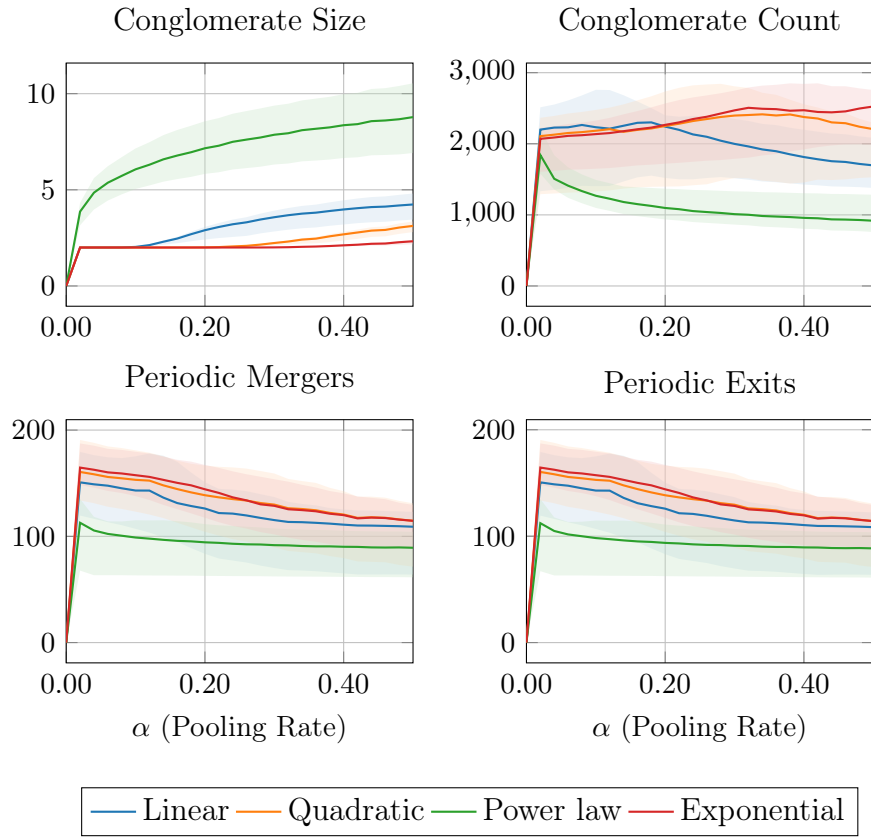


Figure 2: Conglomerate metrics by pooling level across cost structures. The solid line indicates the median and the shaded region is between the 25th and 75th percentile.

To further examine the impact of mergers on the competitive landscape and to deepen our understanding of the role that management cost play, we fit a polynomial of degree two in a panel setting in Equation 10 to the relation between conglomerate size and average market share of firms within a given conglomerate within each simulation. Afterwards we aggregate these results per hyperparameter.

$$\bar{\Omega}_{ct} = \beta_{0t} + \beta_{1t}|O_{ct}| + \beta_{2t}|O_{ct}|^2 \quad (10)$$

We expect the conglomerate size to have a positive impact on the average market share ($\beta_{1t} > 0$). However, we expect management costs to impose a drag high enough to negatively influence the average market share ($\beta_{2t} < 0$). Figure 3 shows that this hypothesis holds only to a certain extent. Initially, for all cost types except the power law one, we see $\beta_1 > 0 > \beta_2$. As we increase the level of pooling, this relation flips, however, the magnitude is much less pronounced, since the scale of the two estimates differs by an order of magnitude. Interestingly, this trend is shifted near identically across cost types, first experienced by the exponential, followed by the linear, and finally by the quadratic type. Again, we notice a pattern where initial sharing increases market share while further increases reverse this effect. For high levels of α all the cost types, bar the quadratic, tend to 0 for both β_1 and β_2 . This is a direct consequence of a decrease in the variability and scale of market share inequality.

Figure 4 provides an initial insight into this dynamic by illustrating the market share distribution for which we collect the quantiles of market shares **within** markets and average over these **across** markets and Monte Carlo simulations and time steps. In our setup with N firms per market a perfectly equal result would assign every firm a market share of $\frac{1}{N}$. In line with expectations, increasing the pooling rate decreases the market share inequality. However, this process is non-linear and proceeds different across quantiles, with market shares not moving at all initially for the largest firms, decreasing at a decelerating rate afterwards. Lower quantiles grow with an increasing pooling rate, while the 99th quantile experiences an initial boost followed by

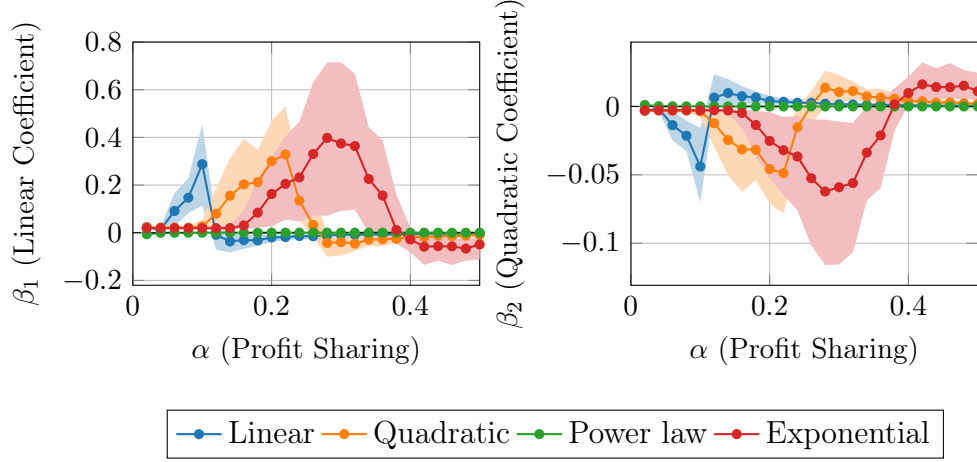


Figure 3: Panel polynomial estimates showing the relationship between conglomerate size and average market share across profit-sharing levels.

a decline. The overall trend is towards more homogeneity in market shares, which explains the decreasing magnitude of the estimates shown in Figure 3, which are heavily influenced by the largest firms. Again, we observe the same pattern, that simulations under the power law cost structure react first, followed by the linear, quadratic, and exponential ones. This reflects the scaling of management cost and the subsequent ability to form larger conglomerates that improve the growth path for each constituent firm in the conglomerate. Ultimately, scenarios with larger conglomerates lead to less inequality. Note that the case of no pooling with $\alpha = 0$ returns the standard result of random multiplicative growth, exhibiting extreme inequality with the largest firm making up almost the entirety of the market and even the 90th percentile dropping to a 0% share [14].

So far we have only looked at a static measure for market shares, meaning that it can only provide information on the distribution of market shares, but not which firms are within them. To address this, we follow each firm throughout a simulation and record both the difference between its maximum and minimum rank and the standard deviation of its ranks. Figure 5 demonstrates how the distribution over both of these metrics changes as we increase the pooling rate α . Note, that given that there are $N \in \{75, 100, 150\}$ firms

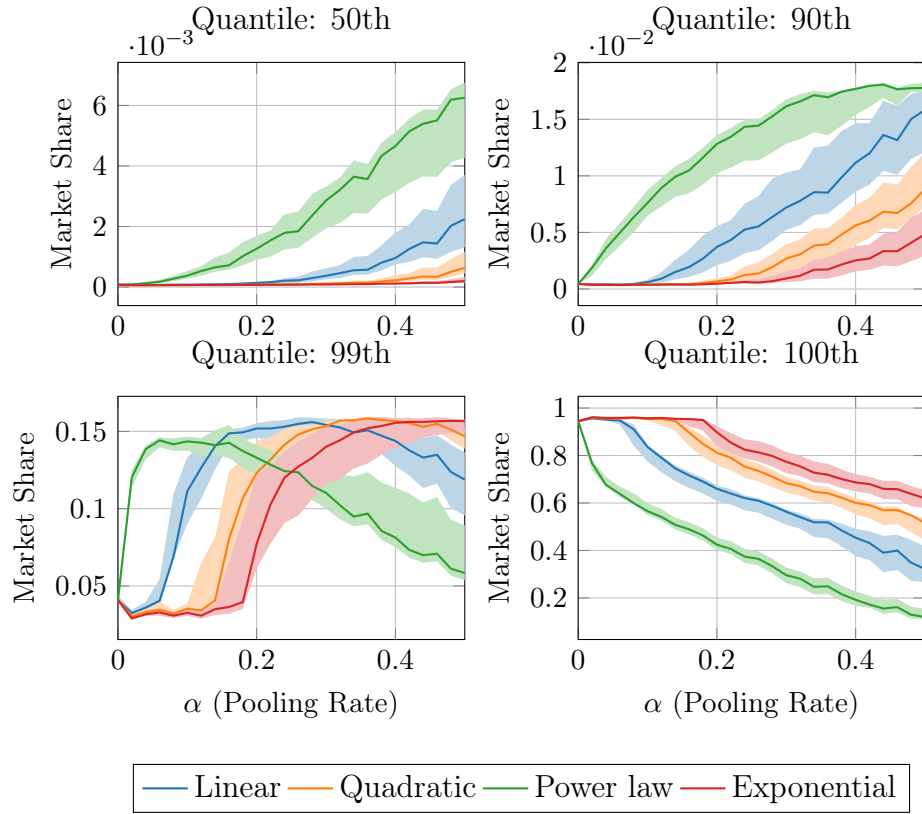


Figure 4: Market share by cooperation level across quantiles and cost structures. The solid line indicates the median and the shaded region is between the 25th and 75th percentile.

per market, there is an upper limit to the limit of the mobility measure. The range shows that even in the no-pooling case the theoretical upper bounds of mobility are reached. However, we observe again that introducing just a small amount of sharing changes the dynamic drastically, as both measures shift completely in their pattern, raising the body of the mobility considerably. In addition, increasing the sharing rate increases the variability of ranks, shifting the distribution to the right. We interpret this as reduced persistence of market positions. The baseline case ($\alpha = 0$) represents pure Gibrat's law dynamics, which are known to produce extreme concentration and lock-in of market leaders [14, 34]. Any deviation from this baseline through pooling therefore represents an improvement in dynamic competition. Note that the theoretical maximum of the range metric (equal to $N - 1$) is reached even at $\alpha = 0$, but the mass of the distribution shifts rightward with increased pooling, indicating that more firms experience rank mobility rather than just a few outliers. Expectedly, there is also variability in the response depending on the cost structure, mirroring the sequence of converge observed above. While simulations based on the power law cost structure reach their maximum mobility for the smallest of $\alpha > 0$, increasing it to higher levels actually decreases mobility slightly, shifting the body of the distribution for both metrics leftwards and increasing spread.

5 Discussion

A recurring theme in the search for determinants of conglomerate formation is diversification to mitigate risks. As specialization of the Smithian [31] and Ricardian [28] nature allows us to increase our efficiency, and as we tend to increase in ever greater manner the degree of roundaboutness of production [37], the potential downsides from miscalculation or shocks looms ever larger. Mitigating these, however, is also becoming simpler [12], as both internal and external means of diversification [15] compete to be the most cost-effective insurance against adverse events. As demonstrated above, true diversification from conglomerate formation offers a reliable mitigation pathway and is applicable to many risk vectors.

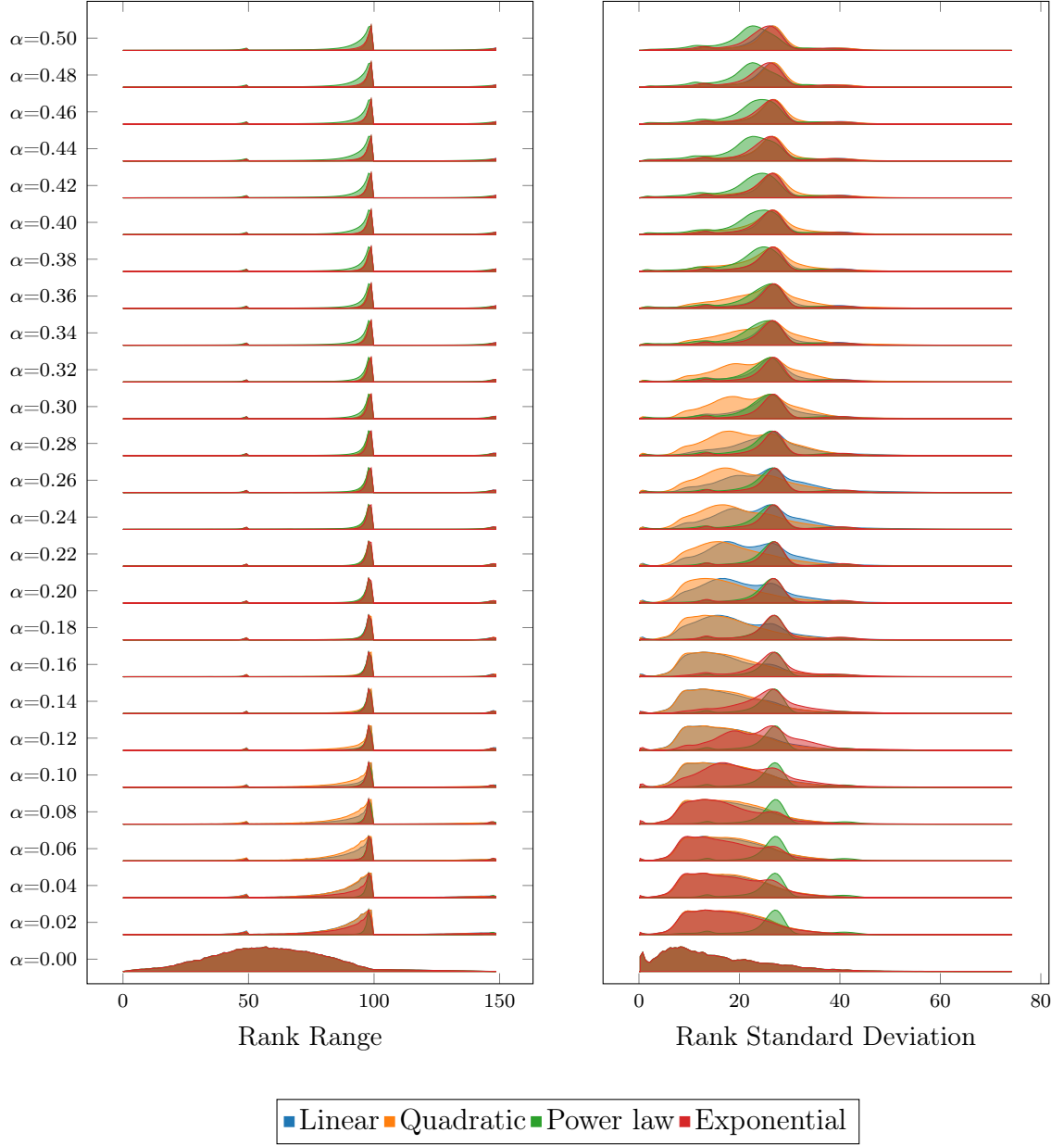


Figure 5: Mobility distributions across profit-sharing levels. Each ridge represents a different α level, with distributions overlaid for all four cost function types. Left panel shows the distribution of rank ranges (maximum minus minimum rank), while the right panel shows rank standard deviation. Higher values indicate greater firm mobility within markets.

Among these are macroeconomic or political shocks that are specific to certain geographic regions or sectors of the economy. The literature on international trade has picked up this theme arguing that cross-country foreign direct investment reduces country-specific risks [26] and even increases between less correlated markets [27]. Similarly, [2] argue that conglomeration can diversify across business cycles, while [3] proposes that it can even be used to match different product life cycles.

Our model assumes independence of growth increments across markets, which maximizes the pooling benefit. In practice, markets exhibit varying degrees of correlation, particularly during economic crises when correlations tend to spike [13]. [24] show that positive correlation reduces but does not eliminate pooling benefits, unless they are perfectly correlated. Our results therefore represent an upper bound on the incentives for conglomerate formation. This has two implications: first, conglomerates will preferentially form across markets with lower correlation structures; second, the competitive effects we identify will be attenuated in highly correlated environments. Testing these predictions empirically—for instance, by examining whether conglomerate mergers are more common across industries with historically lower return correlations—represents a promising avenue for future research.

On the other hand, [18] offers a purely financial rational for conglomeration, where firms merge in order to mitigate financing frictions, and to provide internal capital markets [6]. In addition, pooling can be used to reduce the highly volatile returns on investments that are inherent to R&D efforts, or share the costs associated with them [12].

In fact, many of these effects are already being employed in the businesses seed financing of start-ups, where venture capital firms enforce a low pooling rate for their portfolio companies or those that are part of incubation batches in accelerator programs [7]. Interestingly, this concept is being extended to resources, other than financial, such as physical and human capital, and even reputation. It is thus evident, that pooling can be used to smooth any multiplicative processes, of which there are many that are of interest to economic inquiry [25, 30].

We emphasize that our model speaks to market structure outcomes rather

than welfare. While reduced concentration is often associated with improved consumer welfare in standard models, this connection requires assumptions about pricing, demand, and allocative efficiency that our framework does not incorporate. The welfare implications of the dynamics we identify therefore remain an open question requiring models that integrate our growth-rate-optimization mechanism with explicit market interactions.

6 Conclusion

Our results suggest that conglomerate formation, when combined with resource pooling, may be beneficial not only to the conglomerate and its constituent firms, but also to market dynamics more broadly, decreasing both static and dynamic inequality. We arrived at this result by showing that the standard perspective of the zero-sum nature of resource sharing misses critical details in a world of multiplicative growth. Secondly, we emphasize the difference between portfolio diversification and collaborative resource sharing, allowing us to go further than standard arguments of diversification. We find strong incentives to use conglomerate formation as an insurance mechanism, that if made widely available, would promote competitive dynamics by reducing persistence of market dominance, from which especially new types of business models could benefit.

We see further possible refinements to our approach. Firstly, one could include a pricing mechanism that would allow for more far-reaching statements on the impact on consumers. This would come with a great deal of assumptions such as the nature and interaction of the prices, profits and the underlying growth dynamics. Beyond this, we are aware that the premise of normally distributed growth increments mischaracterizes the process of business growth [17, 33]. In one respect, negative impacts on firm development tend to be heavy-tailed, meaning that impactful events occur much more frequently than normality would predict them [11, 36]. This also holds, though arguably to a lesser degree, for positive impacts, such as the strong growth potential observed in start-ups [22]. Given that large negative developments have an outsized impact on the overall performance, this seems like a strong

use-case for the framework developed herein.

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